

Are Convolutional Neural Networks Biased Towards Texture or Shape? A Layer-wise Investigation of Feature Representations

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Github: <https://github.com/dineshreddymungamuri25-debug/cnn-texture-vs-shape-analysis.git>

1.Introduction:

Convolutional Neural Networks (CNNs) are widely used in computer vision tasks such as image classification and medical imaging. Despite their success, they are often considered “black boxes,” making it difficult to understand how they interpret visual data.

Instead of focusing only on accuracy, this work explores how features are learned across different layers using visualisation techniques. A custom CNN is trained on the Describable Textures Dataset (DTD), and its behaviour is analysed through filters, feature maps, and Grad-CAM. Additionally, a simple experiment is conducted to examine whether the model relies more on texture or shape.

Personal reflection: I initially assumed CNNs would process visual information in a human-like manner — progressing from edges to shapes to objects. However, layer-wise visualisation suggests a different pattern: the network appears to rely more on local patterns across layers, with limited evidence of explicit shape-based representations in this setting.

2.Dataset: Describable Textures Dataset (DTD)

The Describable Textures Dataset (DTD) (Cimpoi et al., 2014) contains 5,640 images across 47 categories representing a wide range of surface patterns, such as bumpy, striped, and cracked. For this tutorial, a subset of 10 classes is selected to reduce training time while maintaining visual diversity.

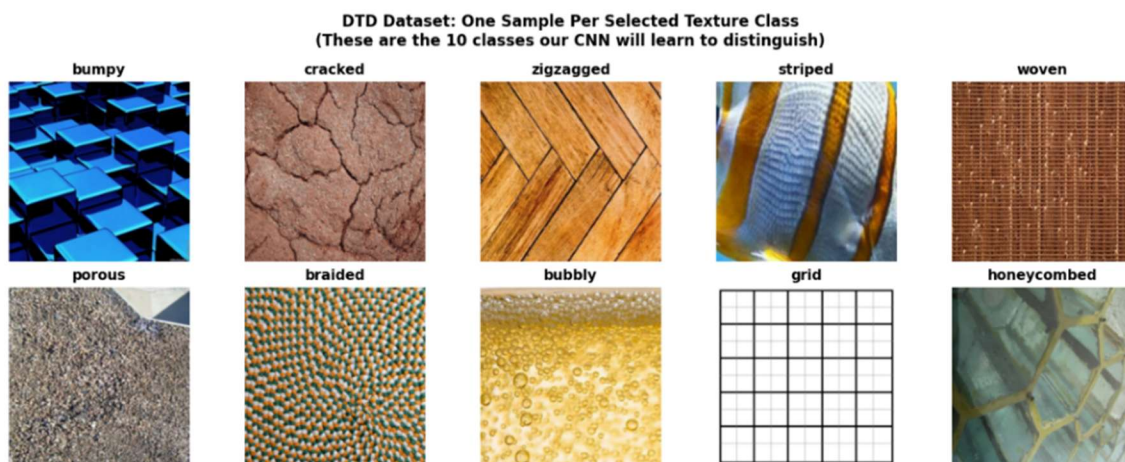


Figure 1. Ten selected DTD texture classes. Chosen for visual diversity — from geometric (grid, honeycombed) to organic (bumpy, porous).
Generated by Cell 4 of the notebook

3.Model Architecture

A custom Convolutional Neural Network (TextureCNN) is designed with four convolutional blocks and trained from scratch. Each block follows the structure:

Each block follows a standard sequence of operations:

Conv2d → BatchNorm2d → ReLU → MaxPool2d

This design allows the model to progressively extract more complex features while reducing spatial dimensions. A custom architecture is used instead of a pretrained model to allow full control and clear inspection of feature learning at each layer.

The model contains approximately 391,000 trainable parameters, making it sufficiently complex for meaningful learning while remaining computationally efficient.

Block	Channels	Output Size	Detects
Block 1 (stage1)	3 → 32	112×112	Edges and colour gradients
Block 2 (stage2)	32 → 64	56×56	Texture primitives (textons)
Block 3 (stage3)	64 → 128	28×28	Repeated structural patterns
Block 4 (stage4)	128 → 256	14×14	Texture class signatures

4.Layer 1 (Block 1) — Edges and Colour

Layer 1 is the first point of interaction between raw pixel data and the network. Its 32 filters operate directly on RGB inputs, allowing them to be visualised as colour patches. This is the only layer where filters can be interpreted directly.

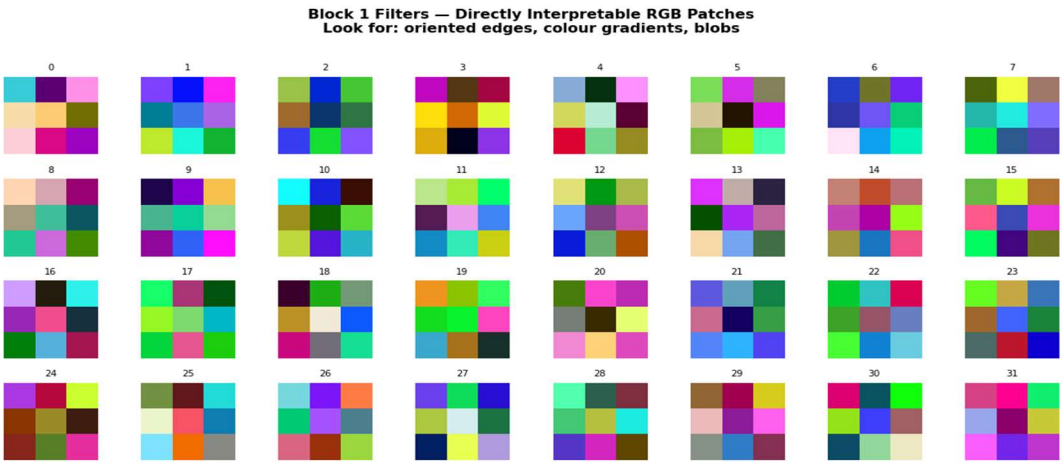


Figure 2. Visualisation of Layer 1 filters showing edges and colour gradients.

The filters capture oriented edges, colour transitions, and simple blobs. These patterns emerge automatically during training and are consistent with findings from Zeiler and Fergus (2014).

Key finding: Layer 1 extracts low-level visual features, including edges and colour gradients, which serve as the basis for hierarchical feature learning in CNNs. These simple features are progressively combined in deeper layers to form more complex patterns and representations.

5. Layer 2 (Block 2) — Texture Primitives

Layer 2 combines the 32 edge maps from Layer 1. Each of its 64 filters operates across all 32 input channels simultaneously — combining edge orientations and colour signals into texture primitives called textons. Because the filters now have 32 input channels, we average across channels to produce a 2D response map for display.

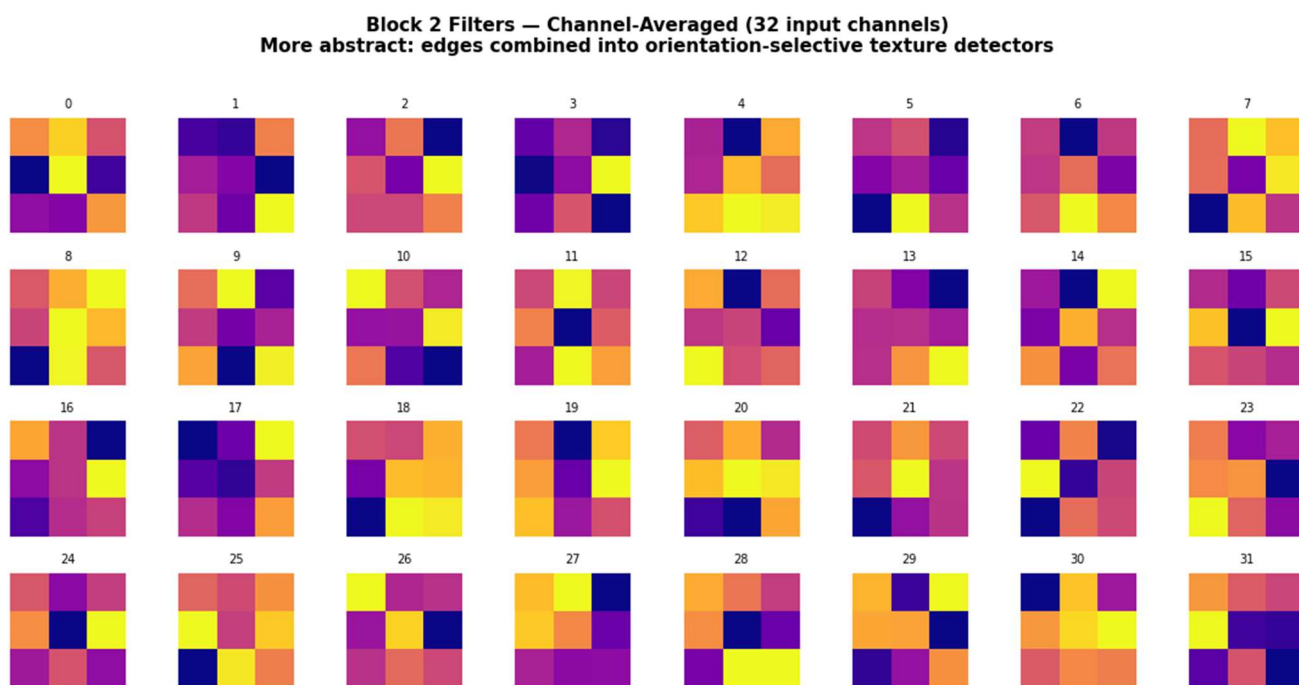


Figure 3. Layer 2 filters (channel-averaged), showing more complex patterns than Layer 1.

Feature maps show that the network begins to detect repeating surface patterns rather than individual edges. For example, in a “bumpy” texture, activations highlight the pattern across the entire image, rather than isolated outlines.

Key finding: Edge features are combined into texture primitives, enabling the model to represent surface appearance rather than simple pixel-level structures.

6. Layer 3 (Block 3) — Structural Patterns

Layer 3 captures how texture elements are spatially arranged. With 128 channels and a 28×28 spatial resolution, each unit now responds to a 32×32 region in the original image. This larger receptive field is necessary for capturing structural repetition across a surface.

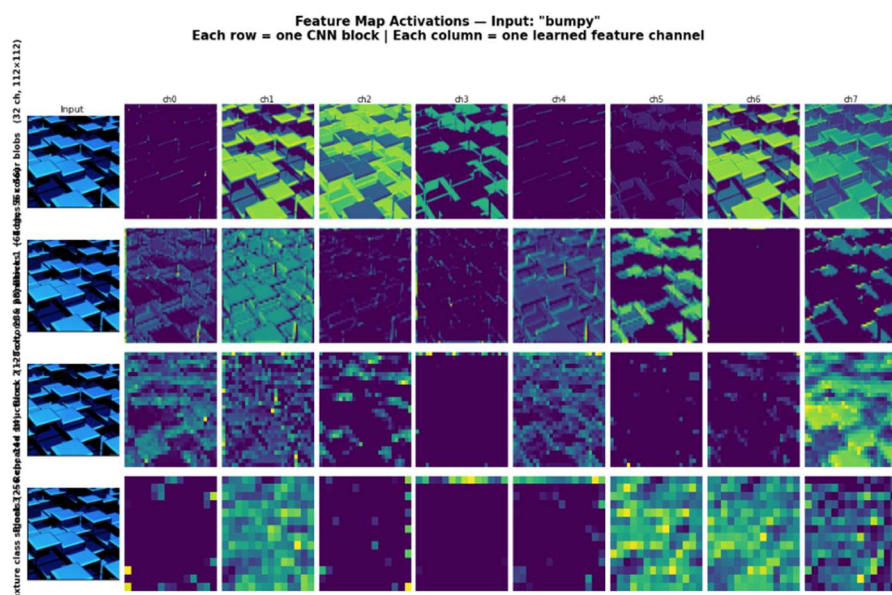


Figure 4. Feature map activations across all four blocks on a 'bumpy' input. Generated by Cell 10. Row 3 shows structural patterns forming at higher abstraction than rows 1 and 2.

At this stage, the model moves beyond detecting individual texture elements and begins recognising how these elements are arranged across the surface. Feature maps highlight repeated patterns over larger regions, indicating that the network is learning spatial relationships rather than isolated features.

The channel activation distribution (Figure 5) shows that Layer 3 is more selective than Layers 1 and 2, with fewer channels activating strongly for a given input. This increased selectivity suggests that the network is developing specialised detectors for different types of structural patterns.

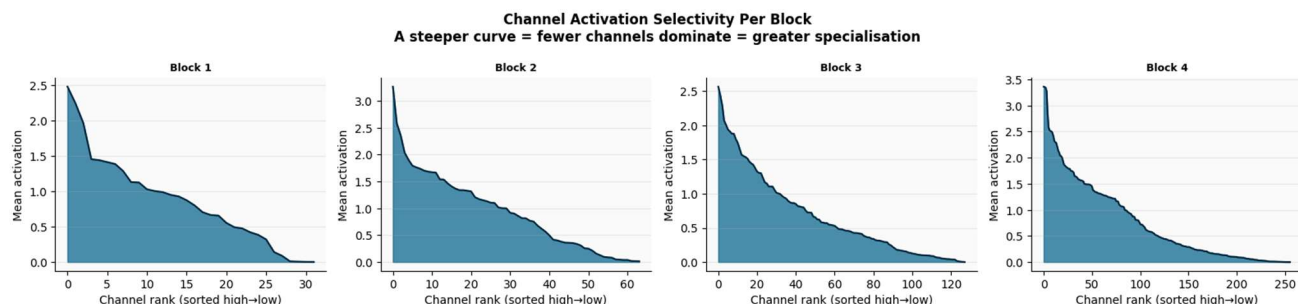


Figure 5. Channel selectivity per layer. The steeper curve at Layer 3 compared to Layers 1 and 2 shows increasing specialisation with depth.

Key finding: Texture primitives are combined into structured patterns that describe how elements repeat and organise across the image. This indicates a shift from local feature detection to more global pattern understanding, with no clear evidence of shape-based representation.

7. Layer 4 (Block 4) — Class Signatures

Layer 4 is the deepest convolutional layer and produces highly abstract feature representations. With 256 channels and a large receptive field, the network captures global texture information across the image, rather than focusing on local details.

Feature maps at this stage (bottom row of Figure 4) appear as diffuse activations with minimal spatial structure. This indicates that the original image has been transformed into a compact representation optimised for classification. Following this, global average pooling reduces the features to a 256-dimensional vector, which is passed to the final classification layer.

At this level, the network no longer represents visible patterns directly, but instead encodes high-level descriptors that summarise the overall texture of the image.

Key finding: The deepest layer produces abstract texture class signatures. At no point in the hierarchy does any layer produce shape-like representations. The network builds a texture hierarchy — not a shape hierarchy.

8. Where Does the CNN Look? Grad-CAM

Gradient-weighted Class Activation Mapping (Grad-CAM) helps identify which regions of an image contribute most to a model's prediction (Selvaraju et al., 2017). It works by backpropagating the class score to a selected convolutional layer and weighting feature maps based on their importance. In this work, Grad-CAM was implemented from scratch to maintain full control over the visualisation process.

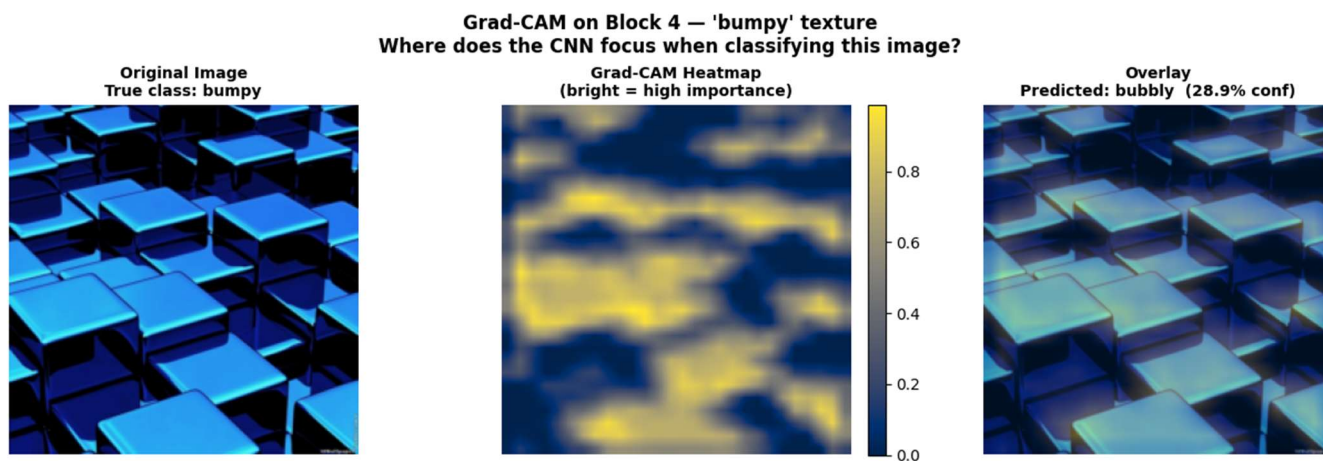


Figure 6. Grad-CAM visualisation for a “bumpy” texture. A colourblind-safe cividis colormap is used, where brighter regions indicate higher importance for the model's prediction.

The heatmap shows that important regions are spread across the entire image rather than concentrated in a specific location.

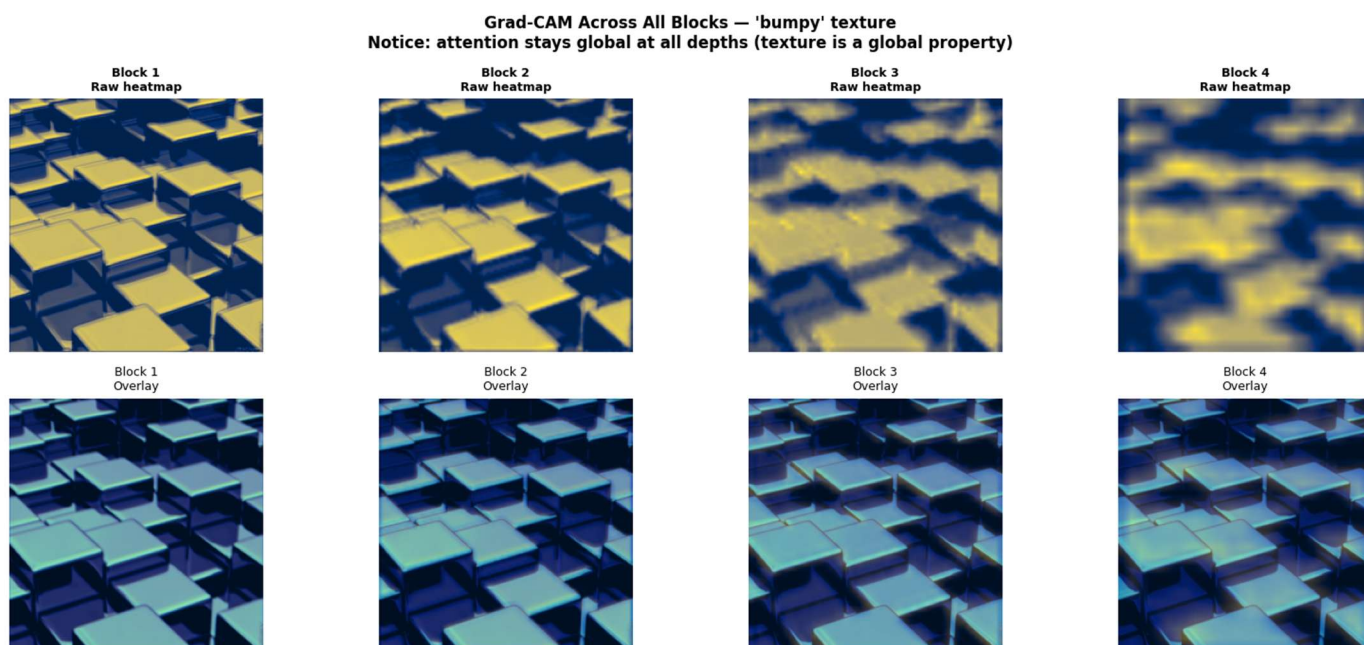


Figure 7. Grad-CAM visualisations across all convolutional blocks, showing how attention evolves with depth.

Across all layers, attention remains widely distributed, indicating that the model relies on global surface information rather than localised features. This behaviour differs from object recognition models, where attention is typically focused on distinct object parts.

Key finding: The CNN distributes its attention across the entire image, suggesting that texture-based classification relies on global patterns rather than specific regions. This supports the idea that texture information is present throughout the image and is used collectively for decision-making.

While Grad-CAM shows where the model focuses, the next step is to test what information it relies on — specifically, whether decisions are driven by texture or shape

9. Texture vs Shape: The Cue Conflict Experiment

Having analysed how features evolve across layers, we now test the central question directly: does the CNN rely more on texture or shape for classification?

To investigate this, texture information is removed using Canny edge detection, producing images that retain only shape outlines. The model's prediction confidence on these shape-only images is then compared to its confidence on the original textured inputs. If the CNN relies primarily on texture, a significant drop in confidence is expected.

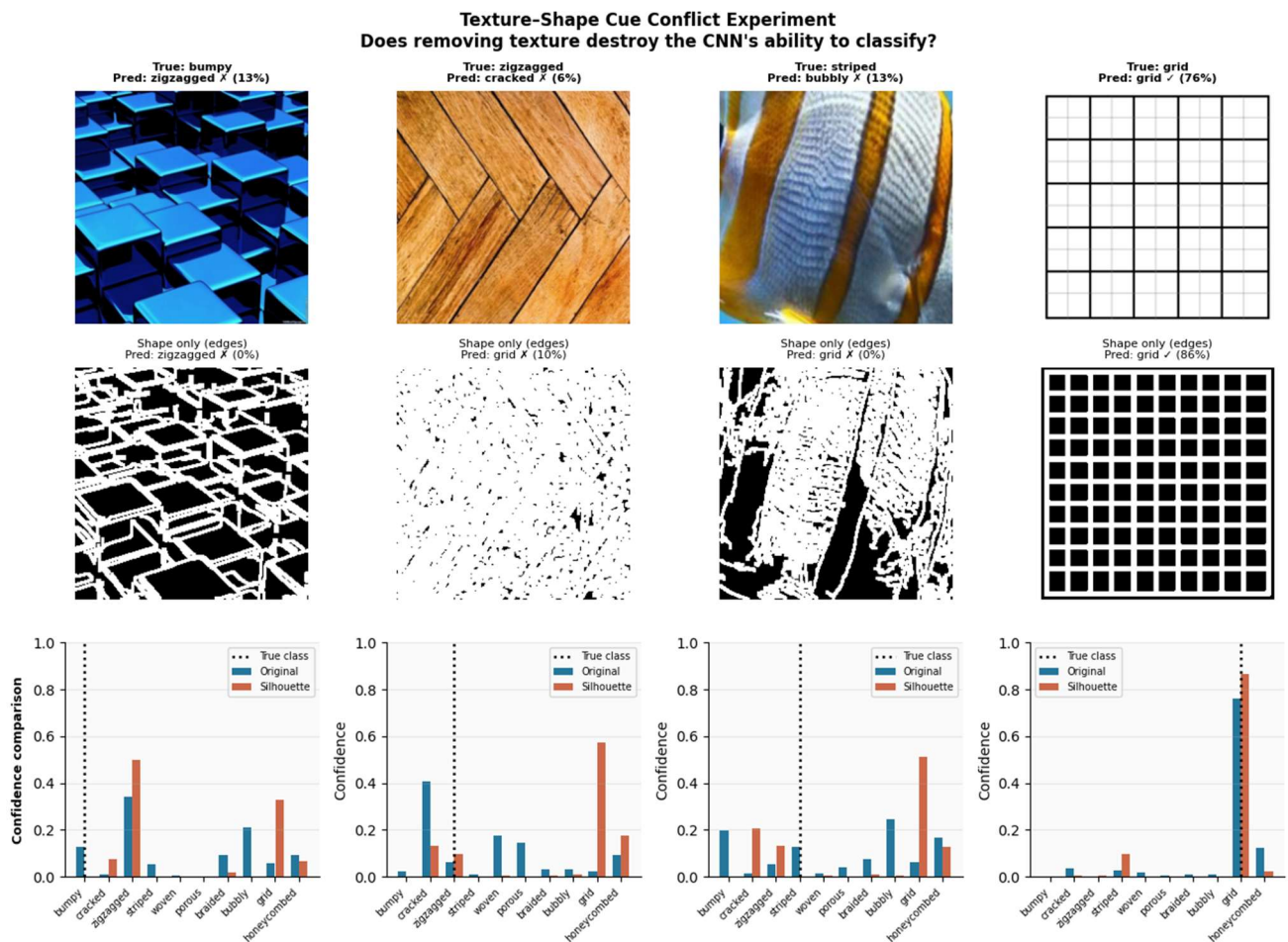


Figure 8. Cue conflict experiment. Top: original images. Middle: shape-only silhouettes. Bottom: confidence comparison.

Results show a clear reduction in confidence when texture is removed. While some classes still produce moderate predictions, overall performance decreases substantially, indicating that texture provides the dominant signal for classification.

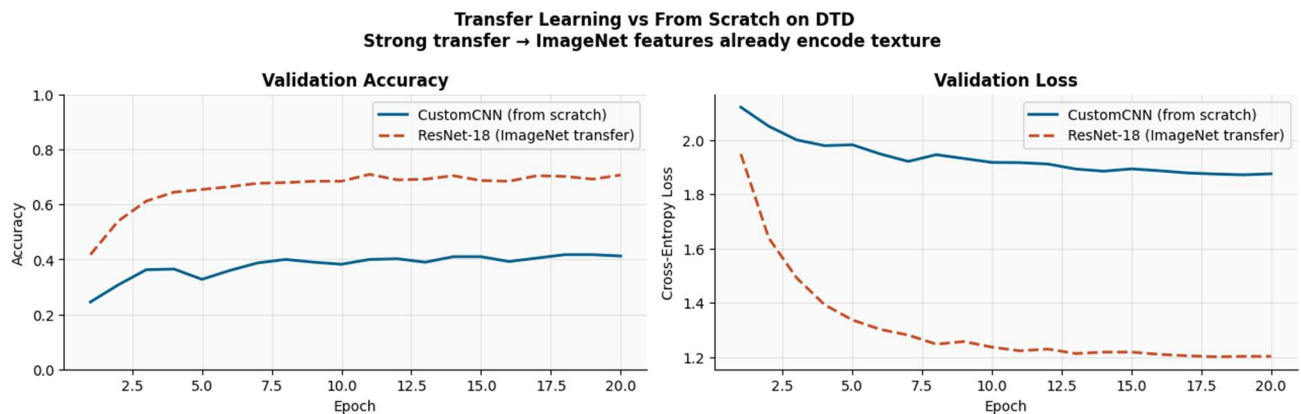


Figure 9. The pretrained model shows faster and stronger performance

To further support this, a comparison is made between a pretrained ResNet-18 model and the custom CNN trained from scratch. The pretrained model achieves significantly higher validation accuracy in early epochs, suggesting that features learned from large-scale datasets already encode useful texture representations.

Experiment result: Removing texture leads to a significant reduction in classification confidence across all tested classes. Although shape information provides some signal, the results suggest that texture plays the dominant role in the model's decision-making process.

10. Conclusion

This tutorial explored how Convolutional Neural Networks learn visual features by analysing each layer individually and testing the role of texture and shape experimentally. Evidence from layer-wise visualisation, Grad-CAM, and the cue conflict experiment consistently highlights a clear pattern in how features are learned.

Final answer: CNNs learn both texture and shape however, texture plays a dominant role throughout the network. Early layers extract basic features such as edges, while deeper layers progressively combine these into texture-based representations. The experimental results further support this observation, showing that removing texture leads to a significant drop in prediction confidence, while shape alone provides limited information.

11. References

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